A newborn gains 2.57 percent of lifetime non-housing, non-drug consumption (Section 6.1). Three quar- ters of it is an externality its parent does not count: 1.50 percent of lifetime consumption that the child gets during childhood, and 0.46 percent from entering adult life without the earnings penalty. The rest, 0.62 percent, has two parts. The first, 0.19 percent, is the part of the child's gains that its parent does count. The second, 0.43 percent, is what the newborn gains later, in its own adult life, like any adult living under Treatment First: a voucher that is easier to draw when sober, and a cheaper drug. The parents lose from the screen's direct impact in partial equilibrium, but once general-equilibrium effects are accounted for, they gain as well, by 0.69 percent of their own lifetime consumption. This is the same adult-life gain as the newborn's 0.43 percent, measured from a different starting point. The parents receive it now, as adults, while the newborn receives it only after its childhood; discounted back to the newborn's birth and set against a lifetime that includes childhood, the gain is a smaller share. In general equilibrium, then, both the child and the parent gain.

The model captures the micro effects that trials measure on the households they follow, and those with a SUD lose from a sobriety condition in the model too. A trial enrols too few households to change the drug price or the chance of drawing a voucher, so the comparison that matches it keeps both at their Housing First levels. Then – percent of the households that have a SUD at the time of the reform lose, and when both adjust, 67 percent still do. The macro effects the trials do not observe outweigh these micro effects: on the households the condition deters from starting to use drugs, on the vouchers its savings buy for others, and on children across the economy.

Even along the transition, homelessness and childhood homelessness do not rise, and children born at the reform gain 1.45 percent of lifetime consumption. A concern, especially among critics of Treatment First, is that homelessness and childhood homelessness could rise, since the households that have a SUD when the reform lands face losing their vouchers unless they enter treatment and stay sober. These outcomes actually fall rather than rise, for three reasons. First, few households lose their vouchers: one with a SUD keeps its voucher by entering treatment and not using, and the screen catches only 20 percent of users a year. Second, drug use falls from the first year, so more households spend on housing instead of drugs. Third, the screen's savings fund vouchers for other households, and reduce homelessness and childhood homelessness among them.

Children born at the reform still gain, but less than the 2.57 percent of those born in the long run, for two reasons. First, more of their parents still have a SUD, since the share with one falls only as fast as recovery allows, and a child whose parent has a SUD enters adult life with the earnings penalty, whether or not the parent is in treatment. Second, they grow up before the chance of drawing a voucher has fully risen, since part of what pays for the extra vouchers is the taxes of adults who grew up without that penalty, and those adults are not yet working.

2 Related literature

Randomised trials of Housing First. A large empirical literature evaluates programmes that provide rapid housing without conditioning on sobriety, and finds large and persistent improvements in housing stability and reductions in shelter and other public-service use (Tsemberis et al., 2004; Padgett et al., 2006; Larimer et al., 2009; Goering et al., 2014;

Aubry et al., 2016). Evidence on substance use is mixed: most studies find no statistically significant change, and some report modest reductions in alcohol-related outcomes. Costantini (2025) finds that enrolling in Housing First lowers the mortality of homeless veterans with mental illness, while the Treatment First programmes of the Department of Veterans Affairs have no lasting effect on it; Section 5.2 compares the model with this finding. The one trial of homeless families, HUD's Family Options Study, followed 2,282 families for three years: an unconditional subsidy reduced later homelessness, family separations and the children's school absences and behaviour problems (Gubits et al., 2016, 2018). These studies, however, measure outcomes over six to thirty-six months, which is short against a recovery process that Vaillant (1995) documents as gradual and protracted. A further limitation of a trial, and perhaps a deeper one, is its scope. A trial randomises households within a fixed environment, and so leaves out exactly the things that decide the welfare comparison: the impact on the behaviour of households outside the sample, the impact on the government's budget, and, most importantly, the effects on the participants' children. No trial of a homelessness programme follows all households, or follows the participants' children into adult life, and no trial randomises what a parent's drug use does to a child at home; that evidence is observational (see the paragraph on intergenerational transmissions below).

Deterrence effects of zero-tolerance policies. The closest evidence on how much a drug test with a severe penalty deters drug use comes from the US military, which in 1981 began testing its personnel at random, 10 to 20 percent of a unit every month, and dismissing those who failed. Mehay and Pacula (1999) use this policy change to compare service members with civilians, following the same people over time. Among those who stayed, past-year marijuana use fell by about 61 percent of what it would otherwise have been; their own estimates, which allow for who left, are smaller. Section 6 uses this response to set how often Treatment First's screen catches a user.

Economic models of drug use. Modelling substance use as forward-looking optimisation dates to Becker and Murphy (1988), and the assumption is empirically defensible at the margin where policy operates. Use responds to its price, even for drugs as habit-forming as heroin and cocaine, and more in the long run than in the short run, the signature of forward-looking choice rather than myopia (van Ours, 1995; Grossman and Chaloupka, 1998; Saffer and Chaloupka, 1999; Caulkins, 2001; Gallet, 2014). Greenwood et al. (2025), who allow perceived and true SUD risk to differ, find that the model-implied beliefs are actually very close to the true risk. Moreover, young people, if anything, overestimate the risk of becoming dependent, and those who perceive a higher risk drink less (Lundborg and Lindgren, 2002). A separate literature emphasises temptation and dynamic inconsistency (Laibson, 1997; Gruber and K˝oszegi, 2001), under which present-biased households over-consume relative to their own long-run preferences. With such assumptions, Treatment First has a rationale even with no externalities at all, since a present-biased household could value a sobriety requirement as a commitment device. That literature is therefore complementary and cuts in the same direction: introducing present bias here would strengthen the case for conditionality rather than weaken it. The paper also departs from Greenwood et al. (2025) in how it generates the income gradient in drug use. There, use concentrates among the less educated because their preferences for drugs differ; here

no preference parameter varies with income, and the gradient comes from the uninsurable

mortality risk of a SUD (Section 3).

General-equilibrium models of homelessness. I build directly on İmrohoroğlu and Zhao (2022), who model homelessness as a rational response to income risk, mental-health shocks, housing costs and housing policy (see also Abramson, 2026, on how eviction policy changes who becomes homeless, and Rotberg and Zhang, 2024, on rent control's effect on homelessness). In that framework housing policy reaches homelessness through income loss and housing costs alone, so a sobriety-contingent policy has no behavioural margin to act on. Adding endogenous use, disorder and recovery, together with state-dependent mortality, creates that margin: the same voucher that insures a household also changes what using costs it, and the disorder it may acquire then feeds back into employment, housing and survival.4

Intergenerational transmissions. The earnings externality rests on what a parent's SUD does to a child. Parental drug abuse is recorded in more than a third of removals into foster care (Adoption and Foster Care Analysis and Reporting System (AFCARS), FY2017), household substance abuse is one of the adverse childhood experiences of Felitti et al. (1998), and children of a parent with a SUD are more likely to acquire one themselves (Mellentin et al., 2016; Yoon et al., 2013). A parent who stays unemployed longer and spends on the drug has less to invest in the child, and that investment shows up in adult earnings: family income raises children's achievement (Dahl and Lochner, 2012), parental investment in education accounts for roughly half of the intergenerational correlation in earnings (Restuccia and Urrutia, 2004), and a year of schooling returns around ten percent (Card, 1999).

Nilsson (2017) is, to my knowledge, the only study that measures, with a natural experiment, how a parental substance shock affects the child's adult earnings. Since this is the best available evidence, the model's productivity penalty is set to its estimate (Section 5), even though the study's shock, exposure to alcohol in the womb, does not map one to one into my model, in which a parent's disorder after the child is born affects the child's productivity. Beyond this study, the only check on the penalty's size is indirect: in the model, a child whose parent has a SUD has odds of acquiring one at the low end of the evidence (Section 5.2), which suggests that the model's penalty is at the low end too.5

As for the study itself, in 1967 and 1968 two Swedish counties let grocery stores sell strong beer for a year. The children conceived in the experiment's first months by mothers under 21 earned 24.4 percent less at age 32 than the cohorts just before and after them and the same cohorts elsewhere in Sweden, a loss that persisted to age 37.

One feature of the estimate makes the calibrated penalty likely conservative, and another makes the estimate the right object. First, the estimate is a group average. Every child conceived by a young mother in the two counties during those months counted as exposed, but only some of those mothers drank more. The average therefore mixes children whose mothers drank more, who lost earnings, with children whose mothers did not, who lost nothing. The loss of a child whose mother did drink more must then be larger than the average of 24.4 percent. Second, it is a loss that lasts, as the model's penalty does. Evidence without an experiment points the same way: adult children of problem drinkers spend more time out of work and earn less (Balsa, 2008). What this paper adds is not another estimate but a price: these effects sit inside an equilibrium model in which the

parent’s use decision is endogenous, so the transmitted loss is a quantity that policy moves rather than a coefficient.

Homelessness persists across generations too (Bassuk et al., 1997). In the model, this persistence comes from the persistence of earnings, not from homelessness itself. The intergenerational elasticity of earnings in the United States runs from about four tenths (Solon, 1992) to about six tenths (Mazumder, 2005), with the same persistence in ranks across the distribution (Chetty et al., 2014). A poor environment does lower a child’s adult earnings beyond that (Chetty et al., 2016; Chyn, 2018), but the model imposes no penalty for growing up in a shelter. However, since Treatment First lowers both homelessness and childhood